

MagicCode Super-Resolution Software API

Reference

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Header: interface/mc_interface.h. v2.2.0 removed mc_enable.h and the MC_Enable helper.

1. Overview

The native API is a C interface for initialization, processing, control, status query, release, and version query. C++ projects include the header normally; symbols use extern "C".

2. Core Workflow

1. Zero-init ctrl_param_t and set input_type, model_path, scaler_factor, alg_mode, backend, and (for temporal) gpu_context plus create-stage depth/HDR. Image size is per frame on mc_nscaler_input_frame_t, not on ctrl_param_t.
2. Create with mc_nscaler_control(&handle, MC_NSCALER_CMD_SET_PARAM, &ctrl, NULL), or let the first mc_nscaler_enable create with defaults (2x SPATIAL_SPEED). Keep the opaque handle.
3. Each frame, fill mc_nscaler_input_frame_t (color + width/height) and mc_nscaler_output_frame_t. Spatial may leave frame NULL. Temporal must set frame to a temporal_frame_t with struct_size, depth, motion, jitter, and camera fields.
4. Call mc_nscaler_enable(&handle, &in_frame, &out_frame).
5. Optionally call mc_nscaler_control with MC_NSCALER_CMD_SET_PARAM or MC_NSCALER_CMD_QUERY_STATUS.
6. Call mc_nscaler_disable exactly once for each successfully created handle.

3. Enumerations

Enum	Values	Notes
mc_cmd_e	MC_NSCALER_CMD_SET_PARAM, MC_NSCALER_CMD_QUERY_STATUS	Control command.
input_type_e	INPUT_BUFFER / R8, INPUT_TEXTURE_RGB8Unorm, INPUT_TEXTURE_R8Unorm, INPUT_BUFFER_RGB	CPU buffers vs GPU textures.
alg_mode_e	SPATIAL_SPEED_MODE, SPATIAL_BALANCED_MODE, TEMPORAL_SPEED_MODE, TEMPORAL_BALANCED_MODE	0..3; MAX_ALG_MODE=4.
magic_backend_e	DEFAULT, X86, NEON, METAL, OPENGL, OPENGLES, VULKAN, D3D11	Runtime selector.
mc_mv_jitter_e	UNSPECIFIED/EXCLUDED/INCLUDED	On temporal_frame_t.mv_jitter.

mc_depth_reversed_e / infinite / hdr	UNSPECIFIED plus conventional/reversed, finite/infinite, LDR/HDR	Create-stage on ctrl_param_t.
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4. Structures

4.1 ctrl_param_t

- No struct_size and no width/height: ctrl_param_t is versioned by MC_CTRL_PARAM_ABI_VERSION; the image size lives on the frame structs.
- scaler_factor spatial [1, 8], temporal (1, 8). Frame width/height [64, 4032] on mc_nscaler_input_frame_t.
- alg_mode: four public modes listed above.
- gpu_context: copied at create; immutable. Vulkan temporal requires physical_device + device. D3D11 temporal requires device. Metal NULL device = system default.
- depth_reversed / depth_infinite / hdr_color: temporal create-stage; ignored for spatial. Immutable after create together with model_path, gpu_context, and backend; SET_PARAM ignores changes to them.

4.2 mc_nscaler_input_frame_t / mc_nscaler_output_frame_t and temporal_frame_t

- mc_nscaler_input_frame_t holds the color resource plus per-frame width/height; mc_nscaler_output_frame_t holds the output resource. Both use the magic_resource_t handle/format/layout/target/mip_count layout.
- mc_nscaler_input_frame_t.frame is NULL for spatial and required for temporal.
- mc_nscaler_input_frame_t.command_buffer: Vulkan temporal recording VkCommandBuffer; optional MTLCommandBuffer on Metal; unused on D3D11/GL/GLES.
- temporal_frame_t.struct_size must cover the v2 prefix through transparency. Tail fields (mv_jitter, exposure_texture) are read only when struct_size covers them.
- Required temporal resources: depth, motion. Optional: reactive, transparency, exposure_texture (1×1 R32F).

4.3 magic_resource_t

handle is a union: pointer (CPU / MTLTexture* / ID3D11Texture2D*), vk_image (VkImage), or gl_texture (GLuint). Exactly one member is live, chosen by the create-time backend. format is backend-native; layout is VkImageLayout (zero for other backends). No width/height: matching handle sizes is a caller contract.

4.4 ctrl_param_t / output_status_params_t

SET_PARAM updates scale and mode on a live handle and ignores immutable fields (model_path, gpu_context, backend, depth_reversed/infinite/hdr_color); size changes arrive through mc_nscaler_enable frame sizes. QUERY_STATUS returns dimensions, scale, mode, backend, gpu_time,

error_code, and when temporal_status_valid is set, the last successful temporal snapshot (scalars only). GPU handles and command_buffer are never stored.

5. Functions

Function	Purpose	Return
int mc_nscaler_enable(void**, mc_nscaler_input_frame_t*, mc_nscaler_output_frame_t*)	Create (*handle NULL) and/or process one frame	0 or negative MC_ERROR_*
int mc_nscaler_control(void**, mc_cmd_e, ctrl_param_t*, output_status_params_t*)	SET_PARAM create/update or QUERY_STATUS	0 or negative
int mc_nscaler_disable(void*)	Release handle	0 or negative
char* mc_nscaler_version(void)	Version string	Static "v2.2.0"

5.1 Vulkan temporal lifecycle

Keep the handle across frames. mc_nscaler_enable only records into the caller-owned recording VkCommandBuffer. Set input color layout, temporal depth/motion.layout, and output layout to the real VkImageLayout at entry (0 is UNDEFINED and fails). The library does not begin, end, submit, or wait. After the caller submits and waits, call mc_nscaler_disable at shutdown — not immediately after mc_nscaler_enable while GPU work may still be in flight.

6. Removed APIs

v2.2.0 removed the v2.0.0 session entry (MC_Init / MC_Process / MC_Uninit / MC_Control) and the v2.1.0 helper (MC_Enable / MC_Disable / MC_Enable_SetModelPath / MC_Enable_SetInputSizeHint) together with mc_enable.h. Use mc_nscaler_enable / mc_nscaler_control / mc_nscaler_disable / mc_nscaler_version. A NULL *handle on mc_nscaler_enable creates a default 2x SPATIAL_SPEED session.

Temporal Super-Resolution

Temporal super-resolution runs through the nscaler session API: libmagic_sr.a (Windows: libmagic_sr.lib) + mc_interface.h + mc_nscaler_enable / mc_nscaler_control / mc_nscaler_disable. v2.2.0 replaced the v2.0.0 MC_Init / MC_Process / MC_Uninit entry and removed the v2.1.0 MC_Enable helper (mc_enable.h is deleted). A NULL *handle on mc_nscaler_enable creates a session with defaults (2x SPATIAL_SPEED) and processes the first frame.

Modes

Value	Enumerator	Role
0	SPATIAL_SPEED_MODE	Spatial, throughput first
1	SPATIAL_BALANCED_MODE	Spatial, quality/cost trade-off
2	TEMPORAL_SPEED_MODE	Temporal, throughput first
3	TEMPORAL_BALANCED_MODE	Temporal, canonical FSR-family path

- Temporal scaler_factor is (1, 8]; 1.0 itself is rejected. Spatial remains [1.0, 8.0].
- TEMPORAL_SPEED_MODE on Metal, Vulkan, and OpenGL ES currently uses a lightweight Convert → Activate → Upscale pipeline. Direct3D 11 and desktop OpenGL 4.3 Speed use the FSR-family temporal path (same family as Balanced, not a separate customer API).
- TEMPORAL_BALANCED_MODE uses the canonical FSR-family path on every GPU temporal backend. Desktop Balanced edge-resolve is supported on Windows Vulkan, Windows Direct3D 11, desktop OpenGL 4.3, and Apple Metal. Android Vulkan and Android OpenGL ES do not run edge-resolve. If enable_sharpening is on, RCAS runs after edge-resolve.
- CPU backends (X86 / NEON) cannot run temporal modes.
- The API accepts (1, 8]. Exact 2× has optimized kernels. Not every scale or device is claimed as device-run.
- Internal pipeline names are implementation detail, not public configuration.

Minimum per-frame inputs

- mc_nscaler_input_frame_t / mc_nscaler_output_frame_t: caller-owned color and output GPU resources.
- mc_nscaler_input_frame_t.frame: non-NULL temporal_frame_t* with struct_size = sizeof(temporal_frame_t).
- temporal_frame_t.depth and motion at input resolution.
- jitter_offset_x/y in input pixels, top-left, +X right, +Y down.
- frame_index / reset_history; camera_near / camera_far / camera_fov_y (radians).
- gpu_context at create (mc_nscaler_control SET_PARAM, or the first mc_nscaler_enable): Vulkan requires physical_device + device; D3D11 requires device; Metal device may be NULL; GL/GLES use the current context (library never makes current).
- Vulkan temporal requires mc_nscaler_input_frame_t.command_buffer to be a recording VkCommandBuffer. Metal command_buffer is optional. D3D11 / GL / GLES leave it NULL.
- Create-stage depth_reversed / depth_infinite / hdr_color on ctrl_param_t (all-zero = conventional-Z, finite far, LDR). Immutable for the handle lifetime.

Motion, depth, and masks

- Motion is current → previous. The library never negates MVs.
- motion_vector_scale_x/y is the only MV numeric conversion. (0,0) defaults to (input_width, input_height). Mixed zero is rejected.
- Depth is GPU device depth in [0, 1]. Linear view-space depth is not accepted.
- reactive / transparency are optional. Explicit caller textures win per channel.
- When omitted, the library derives an approximate mask. That is not a semantic transparency mask.
- Speed on Metal/Vulkan/GLES may derive both missing reactive and an approximate transparency. FSR-family paths (Balanced on all backends; Speed on D3D11/desktop GL) derive missing reactive only and bind an internal zero transparency texture.
- Both temporal modes enable depth+MV history rejection by default.

- Diagnostic only: MAGICSR_TAAU_AUTO_MASK=0 and MAGICSR_TAAU_HIST_REJECT=0 can turn those defaults off. They are not a customer configuration path.

7. Selected temporal error codes

- MC_ERROR_TEMPORAL_NULL_FRAME (-105001)
- MC_ERROR_TEMPORAL_STRUCT_SIZE (-105002)
- MC_ERROR_TEMPORAL_MISSING_COLOR/OUTPUT/DEPTH/MOTION (-105003..-105006)
- MC_ERROR_TEMPORAL_NONFINITE / CAMERA_NEAR_FAR / MV_SCALE (-105007, -105008, -105011)
- MC_ERROR_TEMPORAL_FORMAT (-105015); Vulkan UNDEFINED layout is rejected
- MC_ERROR_TEMPORAL_COMMAND_BUFFER (-105017) Vulkan recording buffer required
- MC_WARNING_TEMPORAL_FRAME_INDEX (105001) not last+1